# Culture consumption and social capital[*]

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**Abstract**

## Introduction

Prevailing models of taste formation, transmission and decay in the sociology of culture posit that current tastes are largely a product of the individual's current network configuration. The standard assumption is that tastes are transmitted through existing network ties, in particular those social connections that are characterized by a high degree of sociodemographic similarity—"status homophily" (Lazarsfeld and Merton 1954)—between ego and alter (McPherson, Smith-Lovin, and Cook 2001). From this perspective, birds of a feather not only flock together, but largely due to this mutual flocking they end up singing, going to the movies, and reading together (Mark 1998b, 2003). A key explanatory strength of the network model is its ability to account for the fact that the distribution of tastes across persons in socio-demographic space is "lumpy" and clustered in proximal regions of this space. The model explains this by suggesting that the dynamics of cultural taste formation and transmission is best envisioned as a "local bandwagon" process of network-based imitation (Mark 2003). Individuals are construed as highly susceptible to the influence of their current contacts and by implication tastes are seen not as enduring aspects of the person but as malleable, "thin" contents liable to change when those network ties in turn change in their turn.

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And change they do. Recent longitudinal network research has shown that rather than being stable over time personal networks as constantly shifting with new contacts constantly being added and deleted throughout the adult life-course (Burt 2000, 2002; Morgan, Neal, and Carde 1997; Suitor, Wellman, and Morgan 1997; Wellman, Wong, Tindall, and Nazer 1997). Thus, what is truly surprising about over-time network “stability” over time is “how *unstable* intimacy is” (Wellman *et al.* 1997: 47). The empirical fact of network volatility coupled with the network assumption of taste as being largely a function of the cultural choices of the individual’s current set of contacts, implies large-scale volatility of those very same cultural tastes. Thus if we take the network perspective seriously, we should expect that as friends change, so do tastes, so we should expect small to moderate correlations between an individual’s taste at time $t$ and her taste at time $t+n$.

Very different assumptions are at core of the cultural capital model of taste formation and acquisition (Bourdieu 1984; Holt 1998; Lizardo 2006). Instead of positing tastes as highly malleable contents at the perennial mercy of a constantly shifting network configuration, cultural tastes are conceived as highly *durable* dispositions toward broad types of cultural objects and performances (Frith 1998). These dispositions are formed in the family and school environment throughout childhood and early to middle adolescence (Smith 1995). Once set, they are highly resistant to change. The reason for this is that as Holt (1997) notes following Bourdieu (1990), tastes are part the person’s *embodied cultural capital* and thus represents a highly resilient aspect of the individual’s relationship to cultural objects (Bourdieu 1984), encoded in the memory traces, semi-conscious affective tendencies and habitual dispositions responsible for practical action (Bourdieu 2000).

Contrary to the network account, according to the cultural capital model tastes should not exhibit over-time volatility, but instead show high inter-temporal correlation. Lizardo (2006) uses this stability assumption to suggest that tastes can in fact be seen as probable (reciprocal) *drivers* of network connections rather than being seen as exclusively *determined* by those connections. Using two-stage instrumental variables and matching methods on cross-sectional data, that study uncovers evidence consistent with this reciprocal effect. Engagement in a wide variety of cultural practices increases network size net of standard controls and after statistically correcting for simultaneity bias. This is in agreement with dynamic theories of cultural and network evolution, which rather than postulating a one-way avenue of causation going from networks to cultural tastes propose a reciprocal (over-time) two-way casual interaction between the two (Carley 1991; Mark 1998a). However, in the aforementioned study the stability hypothesis remains an empirically unverified background assumption. Furthermore to reliance on cross-sectional data, the casual effect of cultural taste on network maintenance or even the more widely accepted effect of network turnover on cultural taste loss, has not been decisively established.

In this paper, I use individual-level longitudinal panel data to (1) empirically adjudicate between these divergent empirical scenarios regarding stability implied by the network and cultural capital models, and to establish (2) direct and unambiguous casual connections between network turnover and cultural taste loss and (3) high rates of cultural activity and the maintenance of high rates of social connectivity in a longitudinal data analysis framework. In the next section, I go on to develop the empirical implications of the network and cultural capital views on more detail. In the following section, I introduce the data and propose my analytic strategy. In the next section, I present the results. I close with a discussion section highlighting the implications of the results for current theoretical understanding of the relationship between cultural tastes and social networks.

## Network versus cultural capital theories of cultural taste loss: some empirical implications

### Network theory

Contemporary network theories of cultural taste and attitude formation and change (and by implication decay) suggest that the contemporary taste profile of an individual is largely a function of the tastes held by the other persons in her surrounding social network (Erickson 1982, 1988). The key idea is that social ties serves as the "conduits" through which new cultural practices are *transmitted* and by way of which individuals *learn* these new cultural practices from their social contacts (Carley 1991; Mark 1998a). This proposition is shared by both straightforward network accounts (i.e. Cardon and Granjon 2005; Erickson 1996; Kane 2004) and more ambitious ecological-network theories of cultural taste (Mark 1998a, 2003; McPherson 2004). Mark (2003: 323-324) for instance speaks of social ties to friends and family as the most important sources of cultural taste acquisition: "[p]eople's friends and family... influence the leisure activities they participate in... participation in various sports, visiting art museums, and gardening are... leisure activities that people often lean from parents, siblings, and/or friends."

Network theories of taste transmission implicitly presume a relatively stable "social structure" coupled to relatively unstable patterns of cultural taste, otherwise it would be hard to think how something that varies at a more rapid pace "causes" or "affects" something that is more stable than itself (Simon 1952). It bears mention that this relative stability assumption is shared by the entire family of network theories that conceive of social structure as the pipes through which contents, or information "flows" and diffuses from one actor to another (Borgatti 2005). This implicit (and crucial) assumption of the stability of social structure, when this term is defined—as it is usually done in network theory—as the concrete patterns of network relations that bind individuals to one another (Berkowitz 1982, 1988; Wellman 1983, 1988; Wellman and Berkowitz 1988), can certainly be questioned in light of recent longitudinal research on constancy of network relations over time. This research—see the 1997 and 2005 special issues of *Social Networks* (volume 19, number 1 and volume 27 number 4) and Burt 2000, for a thorough review)—shows that instead of being constant, network relations are extremely volatile, even when the time frame is constrained to relatively short periods of time, such as a year or less (Bidart and Degenne 2005; Suitor, Wellman, and Morgan 1997).

Most longitudinal studies of personal networks reveal surprisingly high levels of instability in personal networks over time (Lang 2000), with anywhere approximately about 30 to 60% of the personal network experiencing turnover in about a year's time. In his own research on bankers, Burt (2000: 11) finds that "three in four of the 22,709 colleague relations at risk of decay. . . are gone next year." This means that old ties are constantly being deleted—what Burt (2000) refers to as "tie decay"—and new ones formed constantly throughout the life-course (Suitor, Wellman, and Morgan 1997; Wellman *et al.* 1997). Suitor and Keeton (1997: 57), report that more than half of the associates...named at T1 were not named again at either T2 or T3." Morgan et al (1997: 14) conclude that their results "point to a considerable degree of instability in who is present in the network over time. In particular, the average overlap rate of 55% means that just over half of those who were named in one interview or the other appeared in both. Similarly, interpreting the average phi value of 0.34 as a test-retest correlation indicates only a limited ability to predict who was in the network from one time to the next."

These findings imply that for most individuals a recurrent process of new contact selection, extant relationship "fine tuning" and possible deletion of old social ties is constantly at work (Lazarsfeld and Merton 1954). This represents an important—if ill understood—facet underlying the temporal evolution of a person's relational micro-environment. This dynamic acquires renewed salience in light of the observation that recently formed network connections appear to exhibit what Burt (2000), drawing on Stinchcombe (1965: 148-150), refers to as a "liability of newness." This means that dyadic network ties tend to display a high probability of dissolution at early stages of the relationship, with the probability of termination steeply declining as the relationship matures. This proposition implies that the "tie decay function" (that is the probability of a tie being dissolved) is non-linearly decreasing in relationship age, with a high probability for young relationships which decreases as these dyadic ties ripen into close friendships. Burt (2000, 2002) finds evidence consistent with this hypothesis for both strong ties and network bridges (weak ties), respectively.

It is true that network stability is much more substantial in the "core" personal network, with the bulk of the instability concentrated on the periphery (Bidart and Lavenu 2005; Suitor, Wellman, and Morgan 1997; Wellman *et al.* 1997). However, most longitudinal network studies show that the stable "core" network tends to be composed of a small set of alters, primarily kin. The periphery of the network includes most of the persons usually referred to as (non-kin) "friends." Studies show that there is considerable turnover even among those ties usually considered "close friends" (Bidart and Lavenu 2005). It is possible to revise network theories of taste transmission to suggest that taste are mostly transmitted through the small, relatively stable core and not through the larger, unstable periphery. However, most contemporary network theories of taste do not make this distinction, and assume that taste are as equally likely to be learned from family and friends (Mark 2003). It is clear from recent studies on personal communities that the "core" kin network while mobilized for many purposes, including emotional and financial support, is not used for companionship and sociability (Simmel 1949) in any meaningful sense. Most research and theory on culture consumption communities shows that if tastes are exploited in interaction, this is done precisely with those contacts that are usually named as "friends" and which are seen outside the home and the neighborhood (DiMaggio 1987; Fiske 1987; Frith 1998). However, there is nothing inherent in the logic of the network theory of taste transmission that makes it incompatible with current data suggesting large rates of network turnover and the liability of newness of social ties.

Nevertheless, it is important to note that the social transmission hypothesis coupled with the empirical observation of the instability of social ties, jointly imply the inter-temporal instability of cultural tastes. The reason for this is that as Mark (2003) notes a taste that is not "reinforced" by social network influence through interaction is likely to be "forgotten." This is a key assumption of the network based model of taste acquisition. Thus, if we retain the network assumption that individual tastes are largely a function of the tastes displayed by the persons current network alters, and if we accept the fact that these alters are constantly change over the adult life course, then we should expect cultural tastes to themselves exhibit high volatility over time:

*H1: We should observe low to moderate inter-temporal stability of cultural taste across individuals.*

Network theory of course, does not only imply volatile tastes, but directly suggests that any observed amount of taste loss should be primarily associated with life-course events associated with network turnover (i.e. geographical mobility, marriage and divorce, entering and leaving the workforce, adding or dropping social interaction foci, etc.). Surprisingly, but primarily due to the paucity of longitudinal data of cultural taste, this most crucial of assumptions of the network-theoretic view of cultural taste loss has never been assessed in an unambiguous way, although cross-sectional analyses appear to be consistent with computer simulated predictive scenarios (Carley 1991; Mark 2003). The network hypothesis therefore implies that:

*H2: Observed cultural taste loss should be primarily a function of life-course changes associated with network turnover.*

### Cultural capital theory

The cultural capital approach to taste formation provides a very different picture of the stability of tastes than that which can be gleaned from contemporary network theory. Instead of implying that tastes are unstable, "shallow" contents liable to be replaced at the same pace that network relations are replaced, the cultural capital approach suggest that taste are "deep" *durable dispositions and trans-temporally stable forms of embodied cultural capital* (Bourdieu 1986; Holt 1998). It is the relative stability of tastes vis a vis network relations that allows them to function as resources to help *sustain* and *maintain* these network relations over time—primarily by serving as topical resources in conversational interaction rituals (DiMaggio 1987; Lizardo 2006) providing a common focus of attention and affective commitment (Collins 2004). This helps to shift interactions away from purely "instrumental" purposes of information gathering and rational goal attainment to sociable interaction for its own sake (Granovetter 2002; Simmel 1949).

While there has been little empirical research on the dynamic stability of tastes through time, there is good reason to suppose that they are more stable than most network theory leads us to believe. For instance, most studies of the role of early family and school experiences on cultural capital have shown strong effects of arts participation, education and after-school training during adolescence on adult tastes even after controlling for subsequent educational attainment (Kracman 1996). Smith (1995) using data from the 1993 General Social Survey, shows that musical tastes are developed early in young adulthood and appear to be fairly stable across age-cohorts. Holbrook and Schindler (1989) find that persons tend to like styles of popular music with which they were familiar in late-adolescence over older or more contemporary styles, suggesting that tastes tend to be retained over the life-course. Bennett, Emisson and Frow (1999: 178), using qualitative evidence from retrospective interview reports on musical preferences in Australia find that "...many of those who were in middle age had retained the music preference of their late teens and early adulthood." They conclude that while they "...cannot discount the possibility that some people will undergo radical changes in their tastes as they age, on balance the evidence appears to favor the explanation that the taste acquired in the formative years of a cohort persist into later life" (1999: 180). Crook (1997) finds strong correlations of reading and *beaux arts* practices on a three-year panel survey of the Australian population and also across self-reports of cultural activity in adolescence and adulthood, suggesting high stability of cultural practices across the life-course. From this perspective therefore, while the effect of social network change on cultural taste loss is not denied, whatever relational (or "network") effects on taste exist, must operate in the background of high-levels of cross-temporal taste stability and the reciprocal action of cultural taste on social networks:

*H3: We should observe high-levels inter-temporal stability of cultural taste across individuals.*

According to the cultural-capital theory inspired (i.e. Bourdieu 1986) "conversion model" of cultural into social capital (DiMaggio and Mohr 1985; Dimaggio and Useem 1978155; Lizardo 2006), the primary ways in which tastes are used to sustain social networks, is by allowing the individual to connect to alters that are both relationally close—when specialist or "niche" cultural tastes are concerned—and relationally distant—when talking about aptitude for "popular" cultural forms. From this perspective, culture consumption and the ability to decode and appropriate symbolic goods produced in the urban, folk and "culture-industry" realms (Crane 1993) are important resources that individuals mobilize in the *construction* and *maintenance* of active social relationships (Cardon and Granjon 2005; DiMaggio 1987). In this respect the culture conversion model is consistent with Lazarsfeld and Merton's (1954) process model of friendship formation, with similarity based on cultural taste and practices (DiMaggio 1987; Mark 2003) playing the role that "value-homophily" played in the original formulation.[1] As Cardon and Granjon (2005: 302) note in their recent study of the role of culture consumption in the formation of social connections among young adults in France, "...a large proportion of ordinary conversational interaction is generally devoted to comments...on cultural practices and recreational consumption. These are recurrent topics, present in a wide range of social situations...Many social relations are constructed, in different ways, around cultural and recreational activities."

That the origins of new—and reproduction of previously existing—network connections comes from the deployment of cultural knowledge in interaction is implicit in previous models of the relationship between culture and social structure (Fine 1979). For instance, the Carley-Mark (Carley 1991, 1995; Mark 1998a, 2003) "constructural" model can be interpreted as an elementary schema of how culture can be translated into social connections, and how social structure (the distribution of chances to interact across persons in the system) and cultural structure (the distribution of cultural forms across persons) can be defined in an interdependent manner. The constructural model is based on a simple assumption of similarity (homophily), which postulates that the likelihood of a social tie increases with the cultural similarity between any given dyad (a dynamic process similar to the one proposed in Homans 1950 and Lazarsfeld and Merton 1954). In this way the probability that two persons will interact is driven by their cultural similarity. Interaction, in this positive feedback loop, in its turn increases cultural similarity, as individuals exchange their stocks of knowledge with one another.

Thus we should expect that individuals who command a wide variety of tastes which cut across these well defined boundaries (i.e. those who have been referred to as "omnivores" in recent research [Peterson 1992, 1997, 2005]) and who, given the stability assumption, will also be able to "keep" that "tolerant" taste pattern active over time—will also be more effective in sustaining high rates of social

[1] Contemporary network models of taste formation through social network ties (Mark 1998b) can be seen as partially based on the Lazarsfeld-Merton formulation. However, these models raise what was explicitly a secondary dynamic in the original scheme—i.e. mutual "value-adjustment" through social interaction of initially different value dispositions (Lazarsfeld and Merton 1954: 33)—to being the only operative, and thus primary dynamic. It is clear however that for Lazarsfeld and Merton "selective" processes whereby contacts are sifted based on similarity of attitudinal dispositions similar to those emphasized in the culture-conversion account were considered primary (1954: 22-29).

interaction in their—constantly shifting, sparse and loosely bounded—personal communities (Wellman 1979; Wellman, Carrington, and Hall 1988; Wellman and Wortley 1990). This should also help the individual in maintaining ties to important "interaction foci" (Feld 1981, 1982) where old relations are maintained and new social connections are forged.

*H4: High rates of social connectedness at a given time are directly related to cultural taste variety in the previous time period net of reciprocal influences of social connectedness on taste.*

# Data and variables

The data for this study come from the *Project Canada Survey* (PCS). The PCS is a longitudinal study of the social attitudes and cultural practices of a representative sample of Canadian citizens extending from 1975-1995, conducted out of the University of Lethbridge. The project began data collection in 1975 with a total 1,917 respondents. Every five years since representative samples of similar size have been collected, with a "core" set of respondents that have participated in previous surveys. Because it collected data on cultural practices and patterns of social interaction and membership in important foci of social interaction (i.e. voluntary association memberships) over time for the same individuals, the PCS data provide an excellent opportunity to deal with issues of casual order in the relationship between cultural taste loss and life-course events associated with network turnover.

In the following study, I use the two-wave panel covering the years 1985 and 1990. While there is data available for these same respondents for the year 1995, I restrict myself to the core set of respondents who participated in both of these waves (N=560).[2] . I do this for two reasons. First, it is well know that cultural participation declines with age (Smith 1995), so it is advantageous to measure cultural taste with behavioral data before cultural activity begins to tail off. Second, these two waves used a comparable coding on the taste indicators (a slightly different coding was used in 1995) allowing an operationally valid measure of over-time taste loss.

*Data limitations.* While this data provide one of the few existing quantitative sources of information of cultural taste over-time for a large sample of respondents, there are some limitations of the data at hand that deserve mention: first, there are no direct measures of network connectivity (i.e. size of the personal network). The two measures available, frequency of interaction with friends, and number of associational memberships are important components of being sociable but certainly do not represent the notion of social connectivity in its entirety. Second, there are slight changes in the coding of the cultural activity and friendship

[2]This subset of the sample while not representative of the Canadian population can be used to monitor change in habits and practices and the determinants of these changes as it is attempted in the present study. Response rates for 1985 and 1995 where about 60%. About 65% of those interviewed in 1985 where available to be re-interviewed in 1995.

interaction variables across waves, with the introduction of a slightly more elaborate scheme in 1990 (including a "monthly" category) and changing to never category to a composite "hardly ever/never." This may compromise inferences made in investigating taste-loss, by leading to an over-estimation of taste loss events for instance.

I attempted to deal with this last issue by counting a taste loss event as one that involved a shift from one of the extreme categories to the other (the coding of which did not change across waves) and by not using items that showed a clear effect of the shift in wording. For instance, it is clear that for the item related to "going to the movies" something like this happened, with 15% of respondents saying that they "never" go to the movies in 1985, but with 84% saying that they either "hardly ever or never" go to the movies in 1990. A cross-classification of the responses to the same question across time, reveals that this dramatic shift is mostly due to the fact that most of the respondents who answer "seldom" in 1985 (55%), where overwhelmingly more likely to answer "hardly ever or never" in 1990. The same artificial taste shift occurs for other cultural taste indicators associated with *attending events outside the home* (going to a sports event, going to see a dramatic performance in a theater). Cultural practices that do not involve going outside of the home were shown to not be affected by the change in wording, and are therefore the items that I use below in the taste stability and taste loss analyses.

## Variables

### Cultural Taste indicators

*Cultural taste loss indicators.* The PCS collected information for the years 1985 and 1990 related to the frequency of consumption of a variety of leisure culture activities and mass media outlets. To investigate the hypotheses related to cultural taste loss, I selected two activities associated with the consumption of culture and the arts (Alexander 2003; Crane 1993; DiMaggio 2000) as well as one activity related to the consumption of mass print media (van Eijck and van Rees 2000; van Rees, Vermunt, and Verboord 1999) (1) listening to music, (2) reading books of fiction not related to work, (3) reading the newspaper, (4) following a sports team. The criteria for selection of these activities is that they represented common, but relatively low cost (in terms of mobility and geographical accessibility) forms of cultural participation and expression of cultural taste (as opposed to attending a play or going to see a movie in a movie theater, which may decline for factors orthogonal to changing tastes). Respondents were asked to report the frequency with which they engaged in each of these activities using the following prompt:[3]

*Thinking now of your life outside of formal work, how often would you estimate that you: [i.e. listen to music]?*

[3]The various codebooks for all waves of the PCS survey (1975, 1980, 1985, 1990 and 1995) are available online at: http://www.thearda.com/Archive/IntSurveys.asp.

Four categories of responses were allowed: (1) very often, (2) sometimes, (3) Seldom and (4) Never.[4] Thus for each respondent ($i$), for a given cultural activity ($k$) at time ($t$) is defined as their score in the respective ordered categorical variable $y_{ikt.}$

*Cultural taste breadth indicators.* To assess the hypotheses connected to the effects of cultural on indicators of sociability and social connectivity, I created a cultural taste breadth variable for the 1980, 1985 and 1990 waves. This was a simple additive index including participation in a wider range of cultural activities than those used in the cultural taste loss analysis. In addition to the four activities discussed above, the respondent receives one point in the cultural taste breadth scale if he or she reports engaging in the following cultural practices either "very often" or "sometimes": (1) listening to music, (2) going to the movies, (3) going to a theatric performance, (4) going to a sports event outside the home, (45) watching videos at home, (6) engaging in a hobby, (7) reading magazines, (8) following a sports team, and (9) reading a novel or book.

### Network turnover indicators

As noted above, network theory and research provide several guidelines as to what life-course events are associated with radical changes in the personal network. The PCS contains two measures that tap more or less direct transformations of the personal network: (1) longitudinal measures of frequency of interaction with friends and (2) over-time measures of memberships in a variety of voluntary associations (McPherson 1983; McPherson and Smith-Lovin 1987). I also use three other indicators of personal network turnover: (3) changes in educational status (Suitor and Keeton 1997) (4) changes in work status (Fischer and Oliker 1983), (5) changes in geographical location (Degenne and Lebeaux 2005), (6) changes in marital status (Fischer and Oliker 1983) and (7) changes child rearing status (Munch, McPherson, and Smith-Lovin 1997). Since a non-negligible part of the individual's personal network comes from contacts that are often seen at work (Brass 1985; Ibarra 1992; Straits 1996), we should expect that either entrance into or exit from the labor force, or shifts from part-time to full-time work, should have a clear impact on the size and composition of the personal network over time (Bidart and Lavenu 2005). The same can be said of geographical mobility (Relish 1997), with geographically mobile respondents more likely to be subject to radical transformation of their personal networks from those who are not mobile.[5] Finally, a large body of research shows that family status transitions, such as getting married or getting divorced (Gerstel 1988; Leslie and Grady 1985; Milardo 1987; Pahl and Pevalin 2005) and experiencing

[4] As noted above, the 1990 coded the same variables using a slightly more elaborate categorization. I recoded those variables to match the 1985 scheme.

[5] The nominal employment status from which the binary indicator for changes in work status is constructed from has six categories (1) full time, (2) part time, (3) unemployed, (4) retired, (5) keeping house, and (6) other. The geographical mobility item is based on the region in which respondent reported residing in each wave. The marital status variable has five categories: (1) married, (2) cohabiting, (3) divorced, (4) widowed and (5) never married.

widowhood (Anderson 1984; Morgan, Carder, and Neal 1997; Morgan, Neal, and Carde 1997), have significant effects on the social networks of men and women. We should thus expect that transformations in family life, such as getting divorced, getting married or having children (Belsky and Rovine 1984; Bott 1971; Munch, McPherson, and Smith-Lovin 1997), should result in direct effects on the size and composition of the personal network.

### Social connectedness indicators

In order to test the cultural capital hypotheses (3 and 4), I use two indicators of social connectivity in the most recent wave of the survey: (a) rates of social interaction with friends in 1995 and (b) the ability to remain connected to important foci for social interaction from one period to the next, such as voluntary associations. The social interaction measure was obtained from the response to the following question:

*Thinking now of your life outside of formal work, how often would you estimate that you - Spend time with friends?*

In 1985, the allowed responses were: (1) very often, (2) sometimes, (3) Seldom and (4) Never. In 1990, the respondents chose from five response categories: (1) very often, (2) sometimes, (3) Seldom (4) Monthly, (5) Hardly ever or never. Because the models used below are designed to predict interval variables, and due to the fact that there is no reason to expect that the numerical categories used in the survey clearly correspond to the naïve numerical ordering (or that in the case of the categories used in 1990, that the categories are in the *correct* order; is "monthly" more or less that "sometimes") I use association models for cross-classifications (Goodman 1979) to induce the "latent ordering" of the cross-classification of responses to the frequency of interaction with friends across the two waves (Clogg 1982). I use these scores and not the raw categories as the dependent variables in the analyses reported below. The results are shown in Table 1. The table shows that in both 1990 and 1985, the responses do align themselves into a unidimensional continuum (the association model fits the data well; $G^2$=1.29, $p$<0.73), although in 1990, the distance between very often and sometimes is not as large as that between sometimes and seldom.

Retention of connections to important foci interaction is measured using a scale that simply counts the types of voluntary associations in each wave of the survey: these include private clubs, sport clubs, service organizations and hobby clubs. These types of memberships are the kinds of interaction foci in which mass media and arts related culture mediated social interaction is most likely to play a key role in the formation and maintenance of social contacts (Long 2003), and thus on the likelihood that the individual will retain or drop the membership (McPherson, Popielarz, and Drobnic 1992; Popielarz and McPherson 1995).

Figure 1

### Sociodemographic controls

The models shown below include the following sociodemographic controls: gender, age, education, city size, employment status, and marital status, presence of children in the household and city size. The coding and descriptive statistics of all of the variables mentioned above are shown in Table 4 of the Appendix.

## Network driven taste loss

### Analytic Strategy

The cultural taste loss model can be thought of as a discrete time analogue to the usual continuous time event history model for two time periods (Allison 1982), with the dependent variable equaling 1, if a cultural taste loss event is observed and zero otherwise. The coefficients of interest can then be estimated using the following logistic regression equation:

ln (1)

Where $p_{ik}$ is the probability that individual ($i$) reports that he or she "never" or "seldom" engages in cultural activity ($k$) in 1990 after reporting that he or she did so "very often" in 1985 and $M$ is the number of sociodemographic factors included in the equations. The dependent variable is thus our measure that there has been a shift in taste for cultural form $k$ in the 5 year period. $Z$ in (1) is an $i$ x $M$ design matrix of network turnover indicators. The matrix is constructed from each of the sociodemographic and positional variables ($w_{im}$) whose over-time fluctuation (i.e. going from employed in the first time period to retired in the second) is taken to be likely to lead to cultural taste loss The Z matrix is coded according to the following scheme:

$z_{im}^{1985} = 0$ (2)

$z_{im}^{1990} = 0 \quad \leftrightarrow \; w_{im}^{1990} = \; w_{im}^{1985}$ (3)

$z_{im}^{1990} = 1 \quad \leftrightarrow w_{im}^{1990} \neq w_{im}^{1985}$ (4)

Thus for each entry $Z_m$, of the network turnover matrix, a 1 is recorded if a change in sociodemographic status $w$ is observed in 1990 and a zero is recorded otherwise.[6] In the regression model, is thus an $M$ x 1 vector of coefficients associated with the effect of network turnover for $m^{\text{th}}$ covariate on cultural taste loss. If the network hypothesis is correct (hypothesis 2 above), then we should observe >0 for each of the positional shift indicators. Finally, the $X_{ik}$ is equation (1) is a matrix of time invariant (i.e. gender and ethnicity) and time variant (i.e. education and age) individual sociodemographic characteristics and is a $k$ x 1

[6]The voluntary association membership change variable records a one if the individual reports a change in membership status (taken from a binary variable that equals one if the respondent reports belonging to the organization and zero otherwise) for any one of eight types of voluntary associations. Therefore this is a dichotomized version of an originally categorical count variable. The maximum observed number of memberships changes was 6 and the minimum was zero, with a mean of 1.7 and a standard deviation of 1.3.

vector of coefficients corresponding to the effects of these variables on the odds of cultural taste loss.[7]

### Results

*Frequency of Taste Loss.* **Error! Reference source not found.** shows the proportion of respondents who experienced a "taste loss" event from 1985 to 1995. Consistent with the cultural capital hypothesis, across all four cultural taste domains, taste loss is a relatively rare event. This is the case whether this is defined in a "strong" (front "very often" in the first wave to "never" in the second) or in a weaker form (from "very often" to *either* "seldom" or "never"). Less than 1% of respondents go from having a strong taste in listening to music or reading the newspaper to losing that taste in the five year period. A relatively larger number of respondents lose the taste for reading books (1.07%) or following sports (1.43%) but the main empirical finding here is that the great majority of respondents retain their tastes (98%). When taste loss is defined in a weaker form the worst retention rate drops to about 95% for reading books. This supports the cultural capital assumption that tastes are relatively stable over time, a rate of stable that appears to exceed that which can be observed for personal networks (Suitor, Wellman, and Morgan 1997; Wellman *et al.* 1997).

*Taste loss Models.* The coefficient estimates of the discrete-time event history models are shown in Table 2. All of the models have as the dependent variable a taste loss event defined according to the "weak" criterion above. Consistent with the expectations derived from the network-based taste transmission theory (Hypothesis 2), I find that—however rare—the cultural taste loss that can be observed is largely due to positional and social status changes associated with network turnover. Thus, the best predictors of cultural taste loss are changes in the frequency of *social interaction with friends,* changes in the *number of organizational memberships* (${}_k$ >0, $p$<0.05 for *all* cultural activities). Other important determinants of taste loss are status transitions that have been shown in the literature to be clearly associated with network turnover such as changes in educational standing ($p$<0.05 for all cultural activities except reading books and reading magazines) and changes in geographic location ( also statistically significantly different from zero for all four activities). The strongest effects in terms of magnitude, is in accord with the view that tastes change when the personal network is transformed that which pertains to changes in social interaction.

Changes in work status also have the expected positive effects on taste loss, although they are only statistically significant for three out of the four cultural activities (reading the paper is not affected by this factor). The effect of work

[7] Cultural taste and sociodemographic data are available for 560 individuals in the two time periods. Since each respondent yields a case in each of the two panels, the maximum corresponding number of observations for the following analysis is (560 x 2=1120). After deleting observations with missing values we are left with N=1116 in the discrete time event history models.

status on taste for sports in consistent with Erickson's (1996) observation of the key role of "sports culture" in smoothing interaction across class lines in the workplace. The weakest effect of the entire set of network turnover indicators on taste loss belongs pertains to those associated with changes in family status. Changes in marital status is not a useful predictor of taste loss for three out of the four cultural activities, once other sources of network turnover are held constant, although it still predicts changes in taste for listening to music. Changes in the number of children in the household on the other hand do not have a discernible effect on cultural taste loss. Possible reasons for the paucity of effects of changes in marital status include measurement error due to underestimating family volatility (respondents who report being married in both waves are counted as not having experienced a change, even though they might be married to different people or may have experienced other changes in the interim), or the fact that there is little net effect of family status changes left after the primary relational outcomes of this changes (shift in the frequency of interaction with friends) are held constant.

In all looking at the relatively high $^2$ value for overall model fit and the pseudo-R-squared values, it is safe to say that the network hypothesis is strongly confirmed by these data. While the *rates* of cultural taste loss are not consistent with standard network theory, the *sources* of taste loss when it does happen are largely certainly those that we would expect to be responsible for these changes: transformation of the personal network due to life-course changes associated with network turnover.

## Longitudinal Culture conversion model

### Analytic Strategy

In this section I estimate the possible casual effect of cultural capital (as measured by taste diversity) on the maintenance of variables associated with network connectivity over time. There are particular theoretical and empirical challenges in estimating the effects of cultural taste on indicators of social connectivity and frequency of interaction. First, there is the issue of *unobserved individual heterogeneity*: any effect of cultural taste on social interaction observed across panels might be driven by an unobserved, time-invariant individual-level factor that is correlated with both variables. Second there is the issue of *simultaneity bias*: if cultural taste breadth is affected by frequency of social interaction with friends and connectivity to important relational foci, then any estimate of the reciprocal effect of taste on those variables will be biased and inconsistent.

I deal with both of these issues in turn by presenting three different specifications, all of which take advantage of the availability of panel data at two points in time:

*Lagged independent variables model.* First, I present I lagged independent variable model, in which the effects of cultural taste breadth 1985 on measures of social network connectivity in 1990 are estimated holding constant sociodemographic factors measured in 1985. While this strategy takes explicit advantage of the longitudinal aspect of the data, this specification is open to bias produced by any unmeasured, time-invariant individual level factor not included in the model that is constant over time and which thus spuriously produces the cross-temporal association of cultural taste breadth and network connectivity. The longitudinal nature of the data allows us to deal effectively with the issue of unobserved individual heterogeneity by modeling it directly. In the following I opt for a random-effects modeling framework. This model is:

$E(Y_i^{1990}) = a + \gamma C_i^{1985} + \sum \beta x_i^{1985} + e_i$ (5)

Where $Y_i$ is an indicator of social connectivity measured in 1990, $C_i$ is a measure of cultural taste breadth measured in 1985 and x is a vector of sociodemographic control variables measured in 1985.

*Random-effects models.* I also present a model that accounts for unobserved heterogeneity by including an individual level random effect. However, while the random-effects model takes care of the unobserved heterogeneity issue, thus insuring that any observed effect of cultural taste breadth on social connectivity is not spuriously caused by some underlying individual-level factor that is constant across time, it is still open to the simultaneity bias caused by reciprocal causation going from social connectivity to taste breadth (Lizardo 2006). This model can be written as:

$E(Y_{it}) = a + \gamma C_{it} + \sum \beta x_{it} + u_i + e_{it}$ (6)

Where everything is as in (5), except that now the observation pertains to the individual-time period, as shown by the inclusion of the $t$ subscript. This means that each individual yields two observations and the random-effect ($u_i$) is parameterized as being constant across time but varying across individuals. Finally, $e_{it}$ is a classical random disturbance that varies across time and across individuals.

*Instrumental variables random-effects models.* To take this issue into account I also present a third specification, using an *instrumental variables random effects model*. This model accounts for both simultaneity bias and unobserved heterogeneity. I do this by taking advantage of the fact that we have information on the cultural habits and practices of individuals in 1980. I use a lagged version of the cultural taste breadth variable measured in 1980 as an instrument for the same variable measured in both 1985 and 1990 in random-effects panel estimation. Because cultural taste breadth in 1980 is strongly correlated to cultural taste breadth 1985 it is a strong an efficient instrument, allowing us to account for both simultaneity bias and unobserved time-invariant heterogeneity in a longitudinal modeling framework.[8]

[8]This model is the same as (6) except that the measure of cultural taste breadth comes from the *predicted value* of a regression equation that includes all of the control and independent variables and the value of cultural taste breadth in 1980.

## Results

The culture conversion model (Lizardo 2006) implies that individuals who have command of a wide variety of tastes will be able to sustain larger and sparser personal networks over time, allowing them to reap the benefits of those social connections in the form of higher than average rates of social interaction and a higher ability to remain connected to important social foci where new contacts are made and old ones are sustained. The results of the lagged, random effects and instrumental variables regression models of the effect of cultural capital (in the form of "omnivore" taste) in the first time period on network outcomes in the second time period are shown in Table 3.

The results shown in the first model of the table are consistent with the culture conversion model: I find that persons who display a wide variety of tastes in the initial time period sustain higher frequencies of social interaction five years later *even after holding constant individual level predictors of interaction frequency in the first time period* ($>0$, $p<0.05$). While not definitive evidence of "causality" these results serve to cement in a more straightforward longitudinal framework those reported by Lizardo (2006) using two-stage methods on cross-sectional data, thus serving to more firmly establish the primary empirical implication of the culture conversion model: broad cultural tastes are used to sustain larger and in this case more active patterns of interpersonal engagement over time. Furthermore, we can see in the fourth column of the table having a wide repertoire of taste is not only useful in sustaining higher levels informal social connectivity to other members of the personal community network over time, but that it is also helps in raising the chances of remaining tied to important social foci of interaction such as voluntary associations ($>0$, $p<0.05$). As Feld (1981) has noted these interaction foci are important since they serve as the primary social mechanism that allows the individual to form social connections which serve to integrate her into the larger social system. Because recruitment (and the over-time stability of membership) into this foci is largely a matter of keeping alive the network ties that bind the individual to other members of the group (Popielarz and McPherson 1995), it is no surprise that whatever resource helps the individual to sustain social connections to other persons will also appear to sustain his or her connection to these larger social entities.

Are the results from the lagged models the spurious by-product of unmeasured individual heterogeneity or simultaneity bias? The models shown in the second and fifth columns of the table show the results after taking into account unobserved individual heterogeneity by modeling it within a random effects framework. The basic result stands: persons who engage in a wide variety of cultural activities are better able to sustain higher levels of social connectivity both in terms of interaction with members of the "personal community" (Wellman 1979, Wellman et al 1988) and in terms of connection with organizational interaction foci. Furthermore, as shown in the third and sixth columns of the table, the results remain the same (in fact become stronger, suggesting that the simultaneity bias was depressing the magnitude of the cultural taste coefficient downwards) after we deal with both the simultaneity and unobserved heterogeneity bias by using cultural taste breadth in 1980 as an instrument in the random-effects model. Thus we can conclude that in the very same way that being a cultural "omnivore" helps to sustain social connections with friends, cultural variety helps the individual keep their connections to voluntary association foci and sustain high levels of informal sociability with members of the personal community network.

## Discussion and conclusion

This paper advances current theory and research in several ways. First, it provides for the first time conclusive empirical evidence that pertains directly to the empirical adequacy of different assumptions regarding the question of the stability of cultural tastes over time. The question of taste stability is important, since it bears on the direct empirical implications of the two most dominant sociological theoretical frameworks that help us explain the relationship between social ties and cultural resources: network theory whether in its relational (Erickson 1996) or ecological (Mark 1998a, 199b; McPherson 2004) variants, and cultural capital theory (Bourdieu 1984, 1986). Second, this paper produces direct empirical evidence of another crucial assumption of network-theoretic views of taste formation, transmission and decay: that of the effect of relational disruption on taste loss. I tackle this question by attempting to link longitudinal shifts in patterns of social interaction, connection to important social foci (Feld 1981) and life-course transitions associated with network turnover to changes in cultural taste. This represents the first time that this most crucial of implicit assumptions of network theories of taste (Mark 1998b) receives empirical treatment in a research design where issues of casual direction can be dealt with satisfactorily. Finally, this paper builds on previous research on the role of cultural capital and other types of cultural resources on the creation and maintenance of social ties (DiMaggio 1987, 1993; DiMaggio and Mohr 1985; Lizardo 2006; Mische 2003) by providing relatively unambiguous evidence of the reciprocal casual effect of durable patterns of cultural taste on the maintenance of patterns of social interaction and connection to social foci over-time.

The results show that the cultural taste instability corollary of traditional network accounts appears not to be operative in this data. Instead, taste for music, newspaper reading, books, and sports show high degrees of stability, a result consistent with the assumptions of a cultural-capital model. However, even though the taste-stability corollary is not supported by these data, the network theoretic assumption that cultural taste loss is more likely to be due to events associated with the disruption of personal networks receives very strong support. Across all four cultural forms included in the analysis taste loss events are strongly associated with changes in interaction frequency with friends, alteration in patterns of connection to important social foci, and disruptions associated with geographic mobility and changes in work status. Finally, a more stringent test of the cultural conversion hypothesis than that offered by Lizardo (2006) produces robust substantiation for the theory. Even after taking into account both simultaneity and unobserved heterogeneity bias, we find that individuals who command a wide variety of cultural tastes are more likely to sustain higher levels of social interaction with friends and higher levels of connectivity to important interaction settings (such as voluntary association memberships) through the five year period. This effect it is important to underscore appears not to be due to the reciprocal effect of connectivity on cultural taste breadth.

The results presented here have significant implications for how we theorize the statics and dynamics of cultural taste formation, transmission and decay. Rather than the generic idea found in the traditional network model of the individual who is influenced by the tastes of those persons that he or she is connected to, the cultural capital derived hypotheses of relative taste stability and use of cultural taste for the formation of new relationships point to two important additional considerations usually ignored in traditional network-based accounts.

*Choice-homophily of network contacts based on pre-existing cultural dispositions.* Standard network theories of taste transmission draw on the notion of "homophily" (i.e. the empirical generalization that close friends tend to be similar in sociodemographic characteristics) to explain the fact that cultural tastes and practices tend to cluster in certain areas of social space (Mark 2003). From the standard point of view homophilious social connections based on socio-demographic similarity *pre-exist* individual patterns of taste and this explains why cultural practices "lump" in certain areas of social space. That is, the logic of network-based theories leads to them to attempt to explain the fact that cultural practices occupy delimited niches in social space by pointing to two interrelated processes: (1) tastes tend to be associated with social background and (2) individuals tend to associate with those who are socio-demographically similar.

Yet, there is sufficient qualitative evidence (Cardon and Granjon 2005; Long 2003; Wali, Severson, and Longoni 2002)—and as this paper shows, quantitative evidence (see also Lizardo 2006)—to suggest that just in the very same that tastes are transmitted through pre-existing network ties, *new* social ties are formed *on the basis of pre-existing cultural dispositions* (Lazarsfeld and Merton 1954).[9] This is the obverse of the traditional network-theoretic emphasis on the formation of new cultural tastes on the basis of preexisting social ties. In fact, given the relative stability of cultural tastes in comparison to network connections, it is likely that this last process may in fact occur more often that the former process (Lizardo 2006). In this respect (and in contradiction to standard network accounts), the "lumpiness" of cultural tastes in social space may be actually instead be the result of an active—but not necessarily conscious—process of contact selection and relationship fine-tuning *keyed around cultural compatibility* (Lazarsfeld and Merton 1954).

[9] See Illouz 1998 on romantic partnerships and cultural capital DiMaggio 1996 and Ostrower 1998 on taste for the high arts and social relations among elites and Bennett 2004 on taste for popular music and social network formation in online communities.

This proposition is concordant with Burt's (2000) observation of the "liability of newness" of recently formed ties, and with recent studies of changes in social networks of young adolescents and young adults (Bidart and Degenne 2005; Bidart and Lavenu 2005; Cardon and Granjon 2005). Notice that if this is correct, then sociodemographic homophily could in the majority of cases be a *by-product* of cultural compatibility and not the other way around (this cultural compatibility may of course be itself due to socio-demographic of the family of origin [Bourdieu 1984; Illouz 1998]). This is an insight that was always implicit in the theoretical logic of dynamic simulations of network formation based on shared cultural knowledge (Carley 1991), but which has been occluded in recent theory and research due to the one sided emphasis put on the socio-structural determination of shared tastes.

*Class-based heterogeneity in contact choice and structural determination of tastes.* Another insight that is brought about by the current research is the fact that the extent to which contacts are subject to a process of active selection based on taste itself varies according to the individual's command of cultural, educational and socio-economic resources. Put simply, high cultural capital individuals, in particular those employed in prestigious occupations that demand of extensive degrees of cognitive and cultural skill not only have larger and sparser social networks (Lizardo 2006), but the proportion of their networks that is composed of constantly changing, loosely bounded social ties is larger and more heterogeneous than that of culturally disadvantaged individuals employed in routine, low-complexity occupations (Coser 1991; Fischer 1982). In this respect, the endogenous effect of cultural capital in both the ability to select among different social ties and the ability to *reach* across the social structure to form those ties, is itself partially determined by position in social space.

This is important, because the dominant way that "social space" has been conceptualized by network and organizational theorists (i.e. Hannan, Carroll, and Pólos 2003; McPherson 2004) in relation to culture has been—following Blau (1977)—as a multidimensional manifold composed of variables such as age and occupational prestige. The sociodemographic niche for a particular cultural or social form can then be given a metric definition in terms of these variables. Social distance among two points in these space is conceived as being proportional to sociometric distance in social networks, that as being a proxy for the probability that two individuals will interact in real life (Mark 1998a). This presumes that the socio-demographic covariates that define social space are strictly exogenous (i.e. they do not interact in the statistical sense) with the propensity to deploy cultural knowledge reaped from consuming the cultural form in question in interaction to form new social relationships with persons located at relatively distant locations in social space. Traditional network models also assume that there is no interaction between the occupancy of privileged locations in social space and the relative propensity to acquire or learn about specific cultural forms from proximate social contacts. This process is seen as being a generic dynamic applicable to all individuals regardless of their socio-demographic profile.

From the point of view of the culture-conversion model on the other hand the axes of social space as defined above are clearly composed of variables some of which (i.e. educational attainment) themselves are implicated in the extent to which tastes may be driven by social network dynamics. Thus, from this point of view, occupying a privileged position in social space plays an important role in the degree to which individuals are able to loosen the limiting effects of network-based social constraint. This happens due the greater ability of those who posses higher levels of educational and socio-economic resources to deploy a more heterogeneous repertoire of cultural knowledge in everyday interaction. This allows them to form and sustain social relationships that cut-across social dimensions of stratification and differentiation (DiMaggio 1987), as well permitting them to create and maintain intimate ties with those who share their knowledge and tastes. For this reason, the dimensions of "social space" cannot be considered strictly exogenous to the cultural contents whose sociodemographic niche they define. Instead differential command of these cultural contents on the part of individuals may themselves "warp" these dimensions, such that individuals located at some privileged locations in this manifold find it easier to traverse a large distance in that space and thus connect to a wide range of others (Erickson 1996; Lin and Dumin 1986). Individuals located in less privileged areas of the space, on the other hand, are socially and culturally distant from most individuals and cannot connect very well across their locally bounded social circles (Bryson 1997).

There is plenty evidence that is consistent with these considerations. For instance Wellman (1979; Wellman et al 1988; Wellman and Wortley 1990) and Fischer (1982) show that as socio-economic and educational status increases, individuals are much more likely to have role-segmented, specialized relationships in which a clear differentiation is made between those ties designed for sociability and companionship (usually labeled as "friends") and social ties that are the result of non-negotiable statuses such as kinship ties. These lines of research on personal community networks also show that individuals are much more likely to prefer to obtain companionship from voluntary rather than involuntary ties. These are the social contacts that the culture conversion model predicts are likely to be formed and sustained through pre-existing forms of cultural compatibility. These are also the social relationships that are likely to be sustained through interaction rituals in which the symbolic goods coming from artworlds and cultural industries figure as the primary subject of conversation (Cardon and Granjon 2005; DiMaggio 1987). Finally, these are also the same social ties that are continually undergoing high rates of turnover and being constantly recycled and reformed, and which thus require the recurrent deployment of cultural capital in interaction. This goes a long way toward explaining the almost insatiable demand for novel symbolic goods among high cultural capital class fractions (Douglas and Isherwood 1996).

In essence, for a relatively privileged cultural and economic elites, social connections to culturally affine others have come to be disembedded from place, kinship and ascription (DiMaggio and Mohr 1985; Holt 1998), and have taken the form of increasingly longitudinally unstable and thus constantly changing personal community networks (Wellman et al 1988). This increases the chances that differential command of the ability to consume a wide variety of cultural forms becomes a *driver* of the recurrent formation of new network relations. This is less likely to be the case for those individuals who are restricted to a more homogenous, kin-based, and locally constrained set of core alters who change relatively little over time. Consistent with this claim, Lizardo (2006) shows that after holding cultural capital constant, the positive effect of education and socio-economic status on social network size is reduced significantly. This has the curious implication that traditional network theory may more accurately describe the process of cultural taste formation among low cultural capital individuals subject to high levels of network constraint (Burt 2005) while the process of culture conversion may be a more accurate representation of the deployment and operation of cultural taste among culturally advantaged elites.

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Figure

Table 1.

Table 2.

Table 3. Lagged, Random Effects and Instrumental Variables Random Effects Models of the effect of culture consumption diversity on interaction with friends and number of organizational memberships, Project Canada Survey 1985-1990.

| | Interaction with Friends | | | Number of Memberships | | |
|---|---|---|---|---|---|---|
| | Lagged | Random Effects | IV Random Effects | Lagged | Random Effects | IV Random Effects |

| | Interaction with Friends | | | Number of Memberships | | |
|---|---|---|---|---|---|---|
| N. of Cultural Tastes | 0.0289* | 0.119* | 0.142* | 0.0874* | 0.0504* | 0.155* |
| | (3.46) | (17.15) | (5.20) | (3.88) | (4.59) | (2.73) |
| Gender (Women=1) | -0.0623+ | -0.123* | -0.128* | 0.284* | 0.257* | 0.116 |
| | (-1.80) | (-3.16) | (-2.50) | (3.05) | (3.27) | (1.04) |
| Education | 0.0178 | 0.00461 | -0.00680 | 0.0768* | 0.0907* | 0.0671* |
| | (1.60) | (0.38) | (-0.41) | (2.58) | (3.93) | (2.06) |
| Marital Status (Married=1) | -0.0585 | -0.0610 | -0.134* | 0.0572 | 0.00358 | 0.152 |
| | (-1.36) | (-1.31) | (-2.07) | (0.49) | (0.04) | (1.14) |
| Children | 0.00392 | 0.00684 | 0.0155 | 0.0173 | 0.0148 | 0.00837 |
| | (0.36) | (0.55) | (0.92) | (0.59) | (0.59) | (0.23) |
| Age | 0.00486* | 0.000912 | 0.00144 | 0.00194 | 0.00205 | 0.00772+ |
| | (4.35) | (0.76) | (0.78) | (0.64) | (0.87) | (1.66) |
| Urban | -0.00485 | -0.0267 | 0.00220 | -0.115 | -0.0743 | -0.0370 |
| | (-0.17) | (-0.82) | (0.05) | (-1.49) | (-1.13) | (-0.39) |
| Constant | -0.310* | -0.222* | -0.310 | -0.264 | 0.0694 | -0.777 |
| | (-3.41) | (-2.37) | (-1.56) | (-1.08) | (0.38) | (-1.61) |
| $R^2$ | 0.0766 | | | 0.0734 | | |
| $R^2$ (within) | | 0.312 | 0.308 | | 0.00704 | 0.0175 |
| $R^2$ (between) | | 0.123 | 0.133 | | 0.0943 | 0.0703 |
| $^2$ | | 319.7 | 51.55 | | 54.09 | 21.85 |
| N | 560 | 1116[a] | 657[a, b] | 560 | 1116[a] | 657[a, b] |

+p<0.10; *p<0.05; **p<0.01

[a]Unit of analysis is the individual/time-period. Each individual contributes two observations for the analysis (1985, 1990).

[b]Only those individuals with data for three panels (1980, 1985, 1990) can be included in the analysis.

# Appendix

Table 4. Means, standard deviations and range of the variables used in the analysis. Project Canada Survey 1985-1995

| Variable | N | Mean | Std Dev. | Min | Max |
|---|---|---|---|---|---|
| Social Interaction | 561 | 0.81 | -- | 0 | 1 |
| Organizational Memberships | 561 | 0.79 | -- | 0 | 1 |
| Educational Status | 561 | 0.33 | -- | 0 | 1 |
| Geographic Location | 561 | 0.53 | -- | 0 | 1 |
| Work Status | 561 | 0.24 | -- | 0 | 1 |
| Marital Status | 561 | 0.08 | -- | 0 | 1 |
| N. of Children | 561 | 0.08 | -- | 0 | 1 |
| Marital Status in 1985 (Married=1) | 561 | 0.83 | -- | 0 | 1 |
| Marital Status in 1990 (Married=1) | 561 | 0.81 | -- | 0 | 1 |
| Work Status in 1985 (Employed=1) | 561 | 0.65 | -- | 0 | 1 |
| Work Status in 1990 (Employed=1) | 561 | 0.58 | -- | 0 | 1 |
| Gender (Women=1) | 561 | 0.67 | -- | 0 | 1 |
| City Size in 1985 (Large City=1) | 561 | 0.53 | -- | 0 | 1 |
| City Size in 1990 (Large City=1) | 561 | 0.52 | -- | 0 | 1 |
| Education in 1985 | 561 | 2.71 | 1.30 | 0 | 5 |
| Education in 1990 | 561 | 2.85 | 1.29 | 0 | 5 |
| Age in 1985 | 561 | 51.53 | 13.30 | 19 | 87 |
| Age in 1990 | 561 | 55.96 | 14.12 | 24 | 88 |
| N. of Organizational Memberships in 1985 | 561 | 1.97 | 1.40 | 0 | 6 |
| N. of Organizational Memberships in 1990 | 561 | 1.89 | 1.36 | 0 | 9 |
| Freq. of Social Interaction in 1985 | 561 | 1.54 | 0.58 | 1 | 3 |
| Freq. of Social Interaction in 1990 | 561 | 2.78 | 1.02 | 1 | 5 |
| Culture Consumption Scale in 1980 | 339 | 6.76 | 1.19 | 3 | 8 |
| Culture Consumption Scale in 1985 | 560 | 6.43 | 1.72 | 1 | 10 |
| Culture Consumption Scale in in 1990 | 556 | 3.74 | 1.40 | 0 | 8 |

It is well known that measurement error attenuates the relationships between variables. This means that the relationship between two observed variables which are subject to measurement error will generally be weaker than their true relationship. For the analysis of change, this phenomenon implies that when the observed states are subject to measurement error, the strength of the relationships among the true states occupied at two subsequent points in time will be underestimated, or in other words, the amount of change will be overestimated. When the data are subject to measurement error, the observed transitions are, in fact, a mixture of true change and spurious change resulting from measurement error (Hagenaars, 1992; Van de Pol & De Leeuw, 1986). The latent Markov model makes it possible to separate true change and spurious change caused by measurement error (Vermunt et al, 1999: 184).

#

Mixed Markov Latent Class Models

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